JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MAN

KATOL ROAD, NAGPUR

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An Autonomous Institute, with NAAC "A" (

Department of Basic Science & Humanities

Vision	Mission
To lay a robust foundation for the Institute to reach its Zenith.	▪ Achieving academic excellence through rigorous teaching, learning and evaluation practices. ▪ To develop an ability to apply knowledge of basic science and Mathematics to excel in the field of Engineering. ▪ To provide salutary environment for the betterment of faculty and the students.

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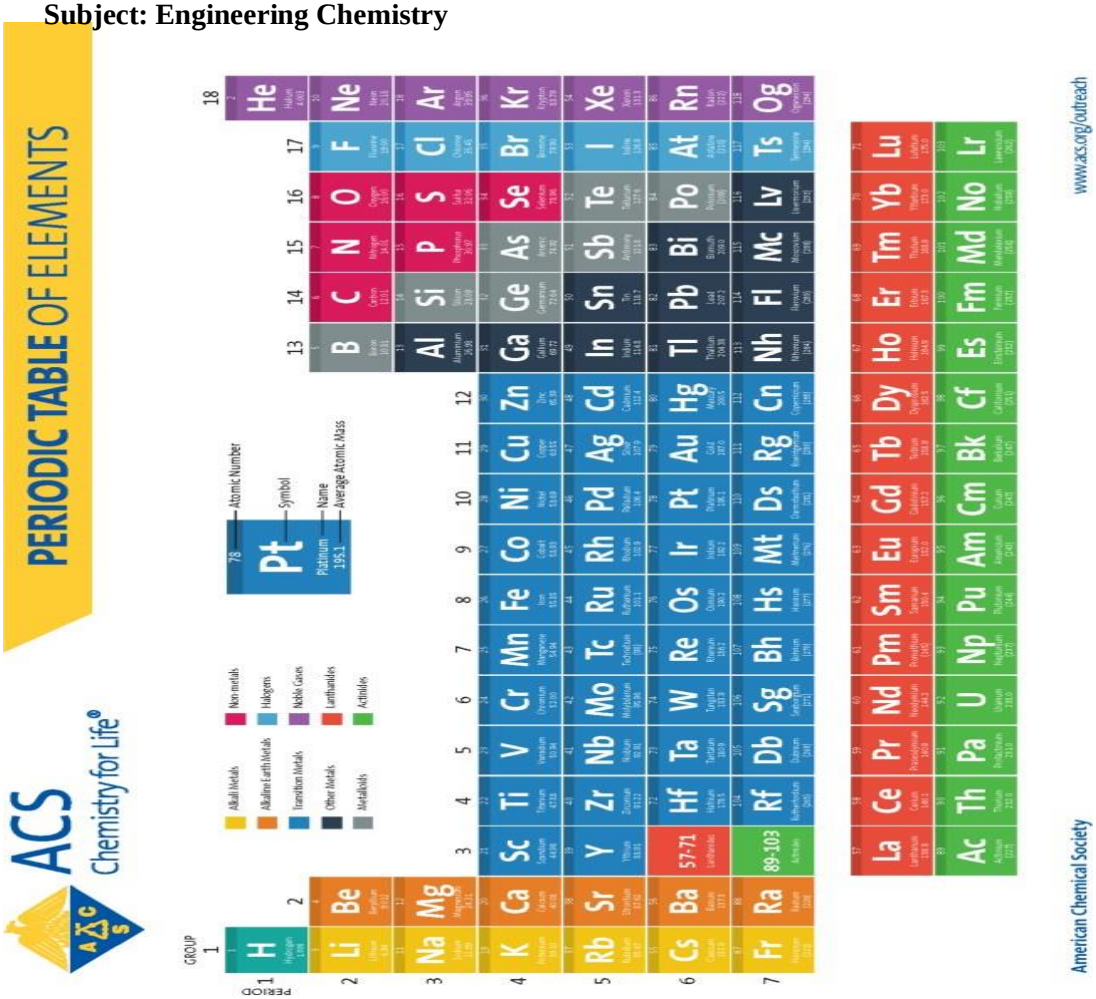


Some of the Theories for Classification of Elements

- **Prout'sHypothesis**

First of all, came the Prout's Hypothesis. Devised in the year 1815, hydrogen wasregardedasthe'central'elementaroundwhichallotheratomsweremade.

- **Dobereiner'sTriads**

This theory came to light in the year 1829. According to the theory, the classification was made in such a way that each group consisted of three elements such that they owned the same properties. Also, in each case, the atomic weight of the middle element is supposed to be the arithmetic mean of the other two elements.

- **Newland'sOctaves**

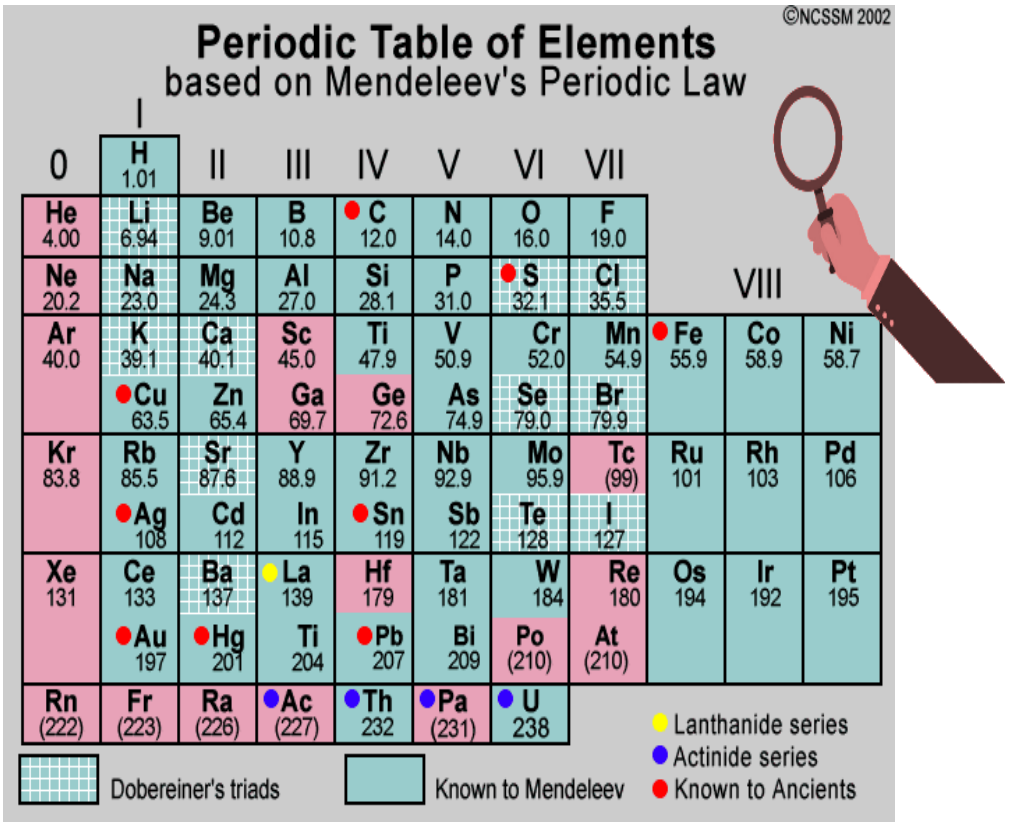
AlsoknownastheLawofOctaves,thistheorywassuggestedintheyear1864.
Newland suggested that elements should be arranged in order of increasing atomic masses. After such arrangement, every eighth element in the arrangement would have same properties as the first element in the arrangement, which follows the order of musicalnotes.

- **Lother Meyer's Atomic VolumeCurve**

The law came in the year 1869 and expressed the existing elements in the form of a curve between the atomic mass of the element and atomic volume. As per this theory, Lother concluded that all elements which hold the same properties came to occupy the same position on the curve.

- **Mendeleev's PeriodicTable**

At the time of Mendeleef, only 63 elements had been discovered. He deduced thatthephysicalandchemicalpropertiesareaperiodicfunctionoftheiratomic masses. Therefore, Mendeleev presented a systematic way to study existing elements.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Periodic Table of Elements
based on Mendeleev's Periodic Law
0 I II III IV V VI VII VIII
He 4.00
H 1.01
Li 6.94
Be 9.01
B 10.8
C 12.0
N 14.0
O 16.0
F 19.0
Ne 20.2
Na 23.0
Mg 24.3
Al 27.0
Si 28.1
P 31.0
S 32.1
Cl 35.5
Ar 40.0
K 39.1
Ca 40.1
Sc 45.0
Ti 47.9
V 50.9
Cr 52.0
Mn 54.9
Fe 55.9
Co 58.9
Ni 58.7
Cu 63.5
Zn 65.4
Ga 69.7
Ge 72.6
As 74.9
Se 79.0
Br 79.9
Kr 83.8
Rb 85.5
Sr 87.6
Y 88.9
Zr 91.2
Nb 92.9
Mo 95.9
Tc (99)
Ru 101
Rh 103
Pd 106
Ag 108
Cd 112
In 115
Sn 119
Sb 122
Te 128
I 127
Xe 131
Ce 133
Ba 137
La 139
Hf 179
Ta 181
W 184
Re 180
Os 194
Ir 192
Pt 195
Au 197
Hg 201
Ti 204
Pb 207
Bi 209
Po (210)
At (210)
Rn (222)
Fr (223)
Ra (226)
Ac (227)
Th 232
Pa (231)
U 238
Lanthanide series
Actinide series
Known to Ancients
Dobereiner's triads
Known to Mendeleev

Modern Periodic Table

In the beginning of the 19th century, a British physicist named ***Henry Moseley*** said that the elements will show periodicity if they are arranged in an order of increasing atomic numbers instead of atomic masses. He believed that the **characteristics of elements are based on their atomic numbers not on their atomic masses.**

Characteristic features of the modern periodic table

- Elements are placed in the increasing order of atomicnumbers.
- There are seven horizontal rows of elements called 7 periods and 18 vertical columns of elements that are called groups. So, there are 18 groups in the modern periodic table that are numbered from 1 to18.
- Thereisadefinitedifferenceintheatomicnumberofelementsinagroup,which is 8, 18, and32.
- Elements are arranged in periods on the basis of number of energy shells in theiratoms.
- First period is the shortest period as it contains only two elements that include Hydrogen (H) and Helium (He).
- The sixth period is the longest period of the modern periodictable.
- The seventh period is still notcomplete.
- The elements that belong to 1, 2, 13, 14, 15, 16, and 17 are known as representative or normalelements.
- Elements from group 3 to 12 are known as transitionelements.
- The inert or noble gases are placed in group18.
- Thef-blockconsistsoftwoseriesofelementswhicharelanthanideandactinide series. The elements of these series or f-block are called rare earth elements or inner transitionelements.
- The elements of group 1 are called alkali metals; of group 2 are called alkaline earth metals; of group 16 are called chalcogens; of group 17 are called halogens and elements of group 18 are called noblegases.
- As we go down a group, the number of shells increases. For example, in the first group hydrogen (H) has one shell, Lithium (Li) has two shells, and sodium (Na) has three shells and so on. However, the valence electrons in the outer shell of all elements in a group as we move down a group remains thesame.
- The valence shell electrons in the valence or last shell of elements increases by one with each step as we move from left to right in aperiod.

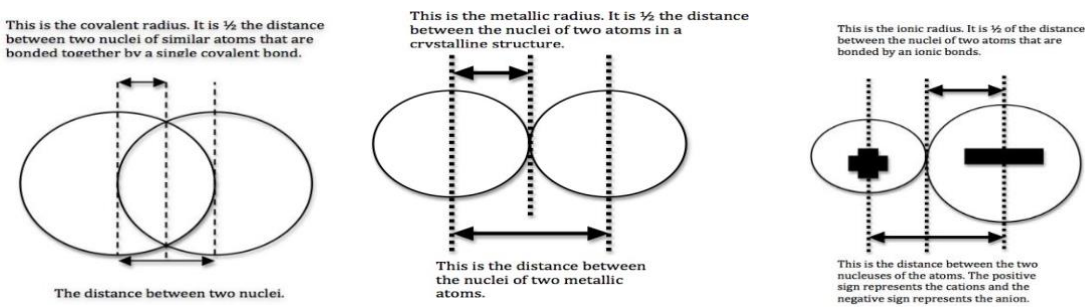
Periodic Properties of the Elements

Theelementsintheperiodictablearearrangedinorderofincreasingatomicnumber. Alloftheseelementsdisplayseveralothertrendsandwecanusetheperiodiclawand table formation to predict their chemical, physical, and atomic properties. Understandingthesetrendsisdonebyanalyzingtheelementselectronconfiguration; allelementspreferanoctetformationandwillgainorloseelectronstoformthatstable configuration.

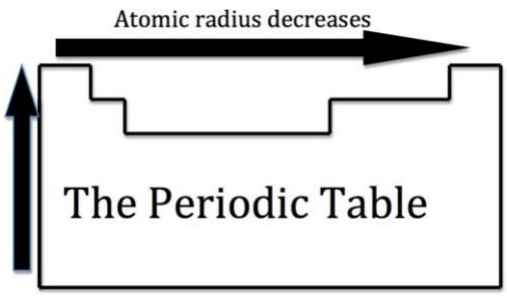
1. AtomicRadius

We can never determine the atomic radius of an atom because there is never a zero probability of finding an electron, and thus never a distinct boundary to the atom. All that we can measure is the distance between two nuclei (internucleardistance).**A covalent radius**isone-halfthedistancebetweenthe nucleioftwoidenticalatoms.**Anionicradius**isone-halfthedistancebetween the nuclei of two ions in an ionic bond. The distance must be apportioned for the smaller cation and larger anion. **A metallic radius** is one-half the distance between the nuclei of two adjacent atoms in a crystalline structure. The noble gases are left out of the trends in atomic radii because there is great debate over the experimental values of their atomicradii.

The SI units for measuring atomic radii are the nanometer (nm) and the picometer (pm). **1 nm = 1 X 10^{-9} m; 1 pm = 1 X 10^{-12} m.**

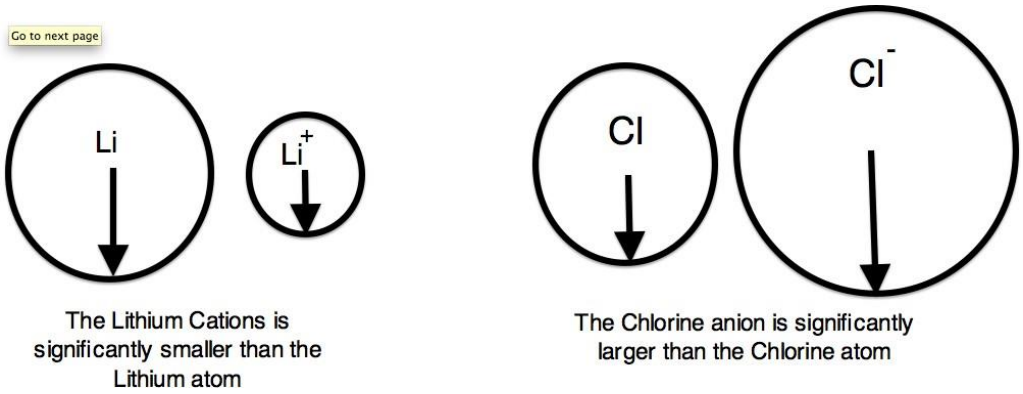


Provided courtesy of Jessica Thornton (UCD)



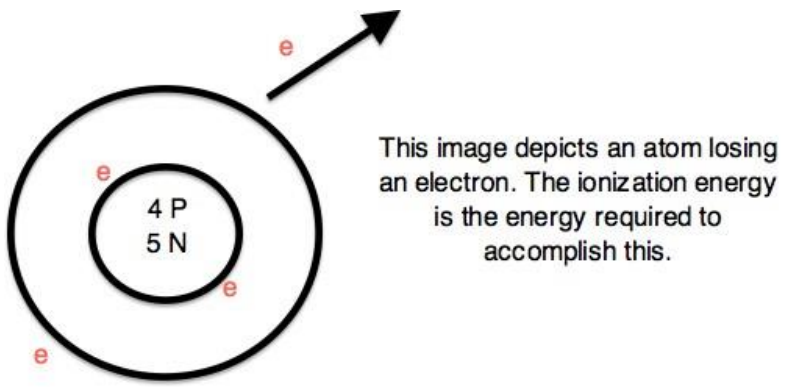
Now we are ready to describe the atomic radius trend in the periodic table. The atomic number increases moving left to right acrossa period and subsequently so does the effective nuclear charge. Therefore, moving left to right across a period the nucleus has a greater pull on the outer electrons and the atomic radii decreases. Moving down a group inthe periodic table, the number of filled electron shells increases. In a group, the valence electronskeepthesameeffectivenuclearcharge,butnowtheorbitalsarefartherfrom the nucleus. Therefore, the nucleus has less of a pull on the outer electrons and the atomic radii arelarger.

Wecannowusetheseconcepttoexplaintheatomicradiusdifferencesofcationsand anions. A cation is an atom that has lost one of its outer electrons. Cations have a smallerradiusthantheatomthattheywereformedfrom.Withthelossofanelectron, the positive nuclear charge out powers the negative charge that the electrons exert. Therefore,thepositivenucleuspullstheelectronstighterandtheradiusissmaller.An anionisanatomthathasgainedanouterelectron.Anionshaveagreaterradiusthan theatomthattheywereformedfrom.Thegainofanelectrondoesnotalterthenuclear charge, but the addition of an electron causes a decrease in the effective nuclear charge. Therefore, the electrons are held more loosely and the atomic radius is increased.



Ionization Energy (ionization potential)

Ionization energy (I.E. or I) is the energy required to completely remove an electron from a gaseous atom or ion. The Ionization Energy is always positive.



The energy required to remove one valence electron is the first ionization energy, the second ionization energy is the energy required to remove a second valence electron, and so on.

- 1st ionization energy

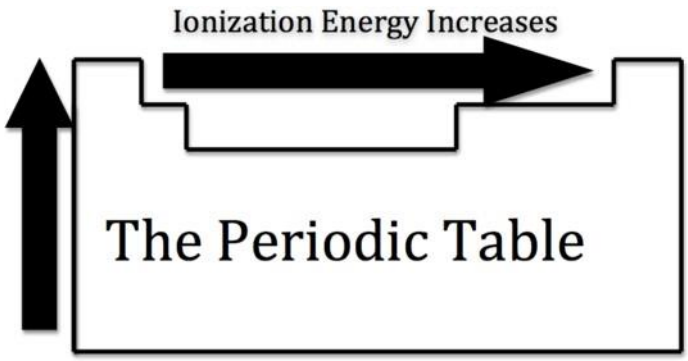
$$Na_{(g)} \rightarrow Na^{+}_{(g)} + e^{-}_{(g)}$$

- 2nd ionization energy

$$Na^{+}_{(g)} \rightarrow Na^{2+}_{(g)} + e^{-}$$

Ionization energies increase relative to high effective charge. The highest ionization energies are the noble gases because they all have high effective charge due to their octet formation and require a high amount of energy to destroy that stable configuration. The highest amount of energy required occurs with the elements in the upperrighthandcorner.Additionally,elementsintheleftcornerhavealowionization energy because losing an electron allows them to have the noble gas configuration. Therefore, it requires less energy to remove one of their valenceelectrons.

Ionization Energies increase going left to right across a period and increase going up a group. As you go up a group, the ionization energy increases, because there are less electron shielding the outer electrons from the pull of the nucleus. Therefore, it requires more energy to out power the nucleus and remove an electron. As we move across the periodic table from left to right, the ionization energy increases, due to the effective nuclear charge increasing. This is because the larger the effective nuclear charge, the stronger the nucleus is holding onto the electron and the more energy it takes to release an electron.



Theionizationenergyisonlyageneralrule.Therearesomeinstanceswhenthistrend does not prove to be correct. These can typically be explained by their electron configuration. For example, Magnesium has a higher ionization energy than Aluminum. Magnesium has an electron configuration of [Ne]$3s^2$. Magnesium has a highionizationenergybecauseithasafilled3sorbitalanditrequiresahigheramount of energy to take an electron from the filledorbital.

Electron Affinity

Electron affinity (E.A.) is the energy change that occurs when an electron is added to a gaseous atom. Electron affinity can further be defined as the enthalpy change that results from the addition of an electron to a gaseous atom. It can be either positive or negative value. The greater the negative value, the more stable the anion is.

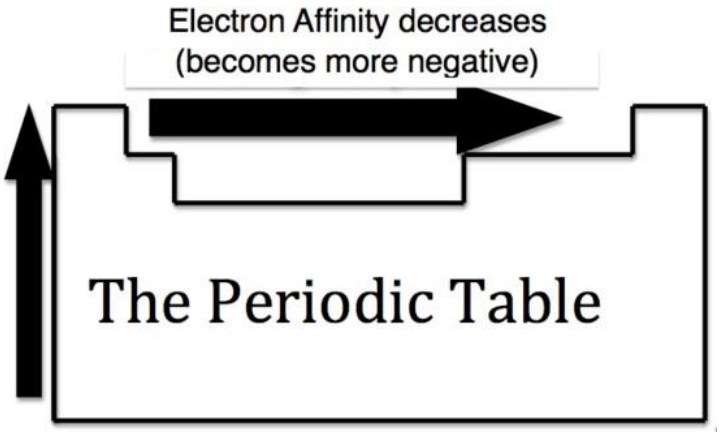
- (Exothermic) The electron affinity ispositive

$$X_{(g)} + e^{-} \rightarrow X^{-} + Energy$$

- (Endothermic) The electron affinity is negative

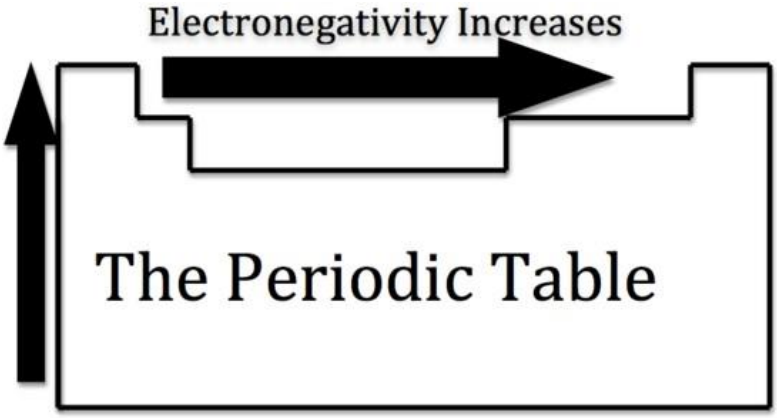
$$X_{(g)} + e^{-} + Energy \rightarrow X^{-}$$

It is more difficult to come up with trends that describe the electron affinity. Generally, the elements on the right side of the periodic table will have large negative electron affinity. The electron affinities will become less negative as you go from the top to the bottom of the periodic table. However, Nitrogen, Oxygen, and Fluorine do not follow thistrend.Thenoblegaselectronconfigurationwillbeclosetozerobecausetheywill not easily gainelectrons.

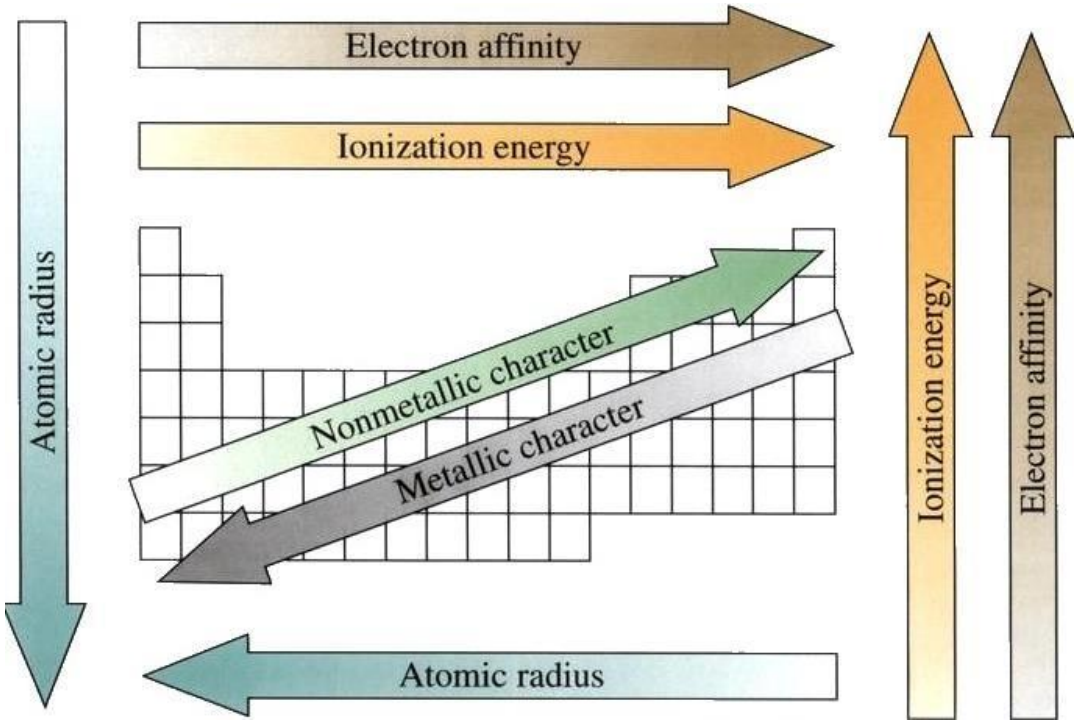


Electronegativity

Electronegativity is the measurement of an atom to compete for electrons in a bond. The higher the electronegativity, the greater its ability to gain electrons in a bond. Electronegativity will be important when we later determine polar and nonpolar molecules. Electronegativity is related with ionization energy and electron affinity. Electrons with low ionization energies have low electronegativities because their nuclei do not exert a strong attractive force on electrons. Elements with high ionization energies have high electronegativities due to the strong pull exerted by the positive nucleus on the negative electrons. Therefore the electronegativity increases from bottom to top and from left toright.



Summary of Periodic Trends



Oxidation State

The oxidation state of an atom is equal to the total number of electrons whichhave been removed from an element (producing a positive oxidation state) or added to an element (producing a negative oxidation state) to reach its presentstate.

- **Oxidation involves an increase in oxidationstate**
  - **Reduction involves a decrease in oxidationstate**

Themoreelectronegativeelementinasubstanceisassigned anegativeoxidation state. The less electronegative element is assigned a positive oxidation state. Remember that electronegativity is greatest at the top-right of the periodic table and decreases toward thebottom-left.

Hard and Soft Acids and Bases (HSAB) Principle

Hard and Soft Acids and Bases (HSAB) Principle is a qualitative concept introduced by Ralph Pearson to explain the stability of metal complexes and the mechanisms of theirreactions..

AccordingtoHSABprinciple,theLewisacidsandbasescanbefurtherdividedint o hard or soft or borderlinetypes.

HardLewisacidsarecharacterizedbysmallionicradii,highpositivecharge,stro ngly solvated, empty orbitals in the valence shell and with high energyLUMOs.

Soft Lewis acids are characterized by large ionic radii, low positive charge, completely filled atomic orbitals and with low energy LUMOs.

Hard Lewis bases are characterized by small ionic radii, strongly solvated, highly electronegative, weakly polarizable and with high energy HOMOs.

SoftLewis bases are characterized by largeionic radii,intermediate electronegativity, highly polarizable and with low energyHOMOs.

The **Borderline** Lewis acids and bases have intermediate properties.

Remember that it is not necessary for Lewis acid or base to possess all the properties to be classified as hard or soft or borderline.

In short, Hard acids and bases are small and non-polarizable, whereas Soft acids and bases are larger and more polarizable.